HAPPIEST HEALTH | Happiest Health Digital Twin CHALLENGE

A SYNTHETIC RESEARCH PROTOTYPE
Not for clinical use or diagnosis

GLYCEMIC TWIN

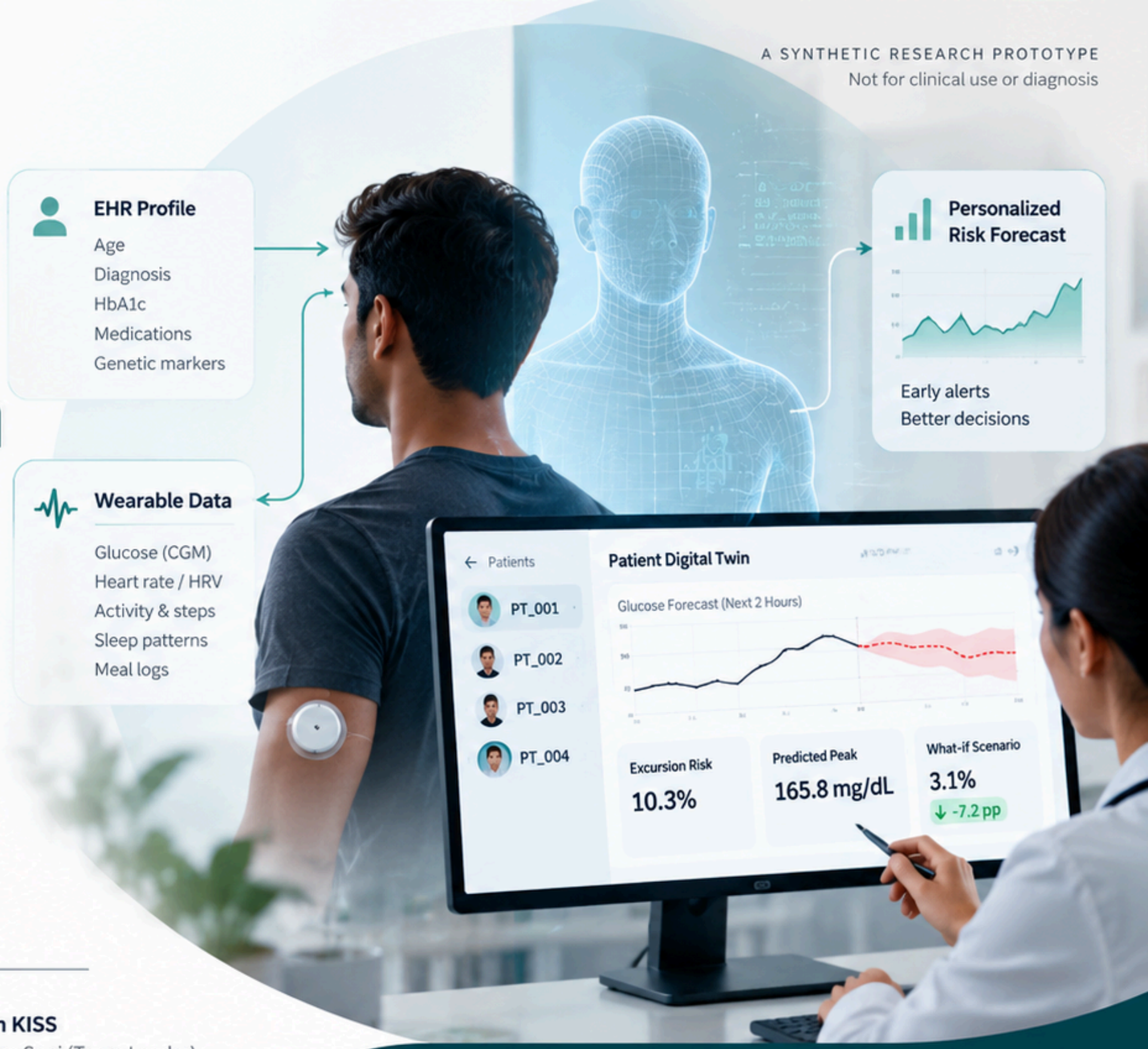
Patient Digital Twin

A synthetic-data-driven glucose forecasting and clinical decision-support prototype.

Historical EHR Data
Clinical profile, labs, history

Real-time Wearable Data
Glucose, activity, sleep, heart rate

AI-Powered Forecasts
Early risk insights, what-if scenarios



Digital Twin Challenge 2026
Happiest Health

Indian Institute of Management Jammu

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PROACTIVE HEALTHCARE FOR A HEALTIER INDIA

PROBLEM STATEMENT

From fragmented data to earlier warning

Today, critical information about a patient's health lives in separate systems.
A single data source does not provide a complete picture, making it difficult to anticipate a glucose excursion before it happens.

Historical Clinical Records (EHR)

Explain the patient's underlying risk

- **Demographics** (Age, Gender)
- **Diagnoses** (Type 2 Diabetes, Comorbidities)
- **Lab Results** (HbA1c, C-Peptide, eGFR)
- **Medications** (Metformin, Insulin, etc.)
- **Genetic Markers** (TCF7L2 risk)

Real-Time Wearable / IoT Data

Show what is happening now

- **Glucose (CGM)** (Every 15 minutes)
- **Heart Rate / HRV** (Activity, stress patterns)
- **Activity & Steps** (Physical movement, kcal)
- **Sleep Patterns** (Duration, quality, stages)
- **Meal Logs** (Timing, carbohydrate intake)



Can we combine patient history and changing signals to forecast a potential glucose excursion before it occurs?

Enable earlier alerts and better-informed clinical decisions

USERS

Who benefits from the Digital Twin?

A shared view of a patient's historical profile and real-time signals helps doctors make more informed, proactive and personalised care decisions.

1 Physician
Primary user



- Review patient profile and risk forecast
- Inspect glucose trends and signal quality
- Explore what-if scenarios
- Get interpretable insights for decision support

2 Patient
Secondary user



- Understand personal health trends
- Receive timely alerts (with clinician guidance)
- See impact of lifestyle changes (e.g., activity, diet)
- Stay engaged in long-term self-care

3 Care Team
Operational user



- Monitor multiple patients
- Identify high-risk individuals
- Review alerts and outcomes
- Support coordinated care plans

Clinical review & decision support

Informed and empowered patient

Coordinated and proactive care

Patient Digital Twin

Unified view · Real-time signals · Predictive insights

USER PAIN POINTS

Why ordinary monitoring is insufficient?

Doctors and patients have access to more data than ever, but key information remains fragmented, hard to interpret and not **forward-looking**, making it difficult to prevent glucose excursions.

1 Fragmentation of data

- **EHR** (HbA1c, labs, diagnoses)
- **Wearables** (CGM, HR, sleep, activity)
- **Medications** (insulin, metformin, etc.)
- **Genetic &** ather risk factors (e.g., TCF7L2)

Patient information sits in separate systems with no unified, real-time view.

2 Limited anticipation



A current reading does not show the full future trajectory, making it hard to act early.

3 Alert uncertainty

High Glucose Alert
2:00 PM

- Is this a real risk or sensor noise?
- Poor-quality or missing data
- Frequent false alarms
- Hard to prioritise patients

Sensor errors and data gaps can trigger unreliable alerts and create alert fatigue.

4 Limited interpretability

Risk score: 0.52

Why is the risk high?

- ?
- ?
- ?

A risk score without clear contributing factors is difficult to interpret and act on.



Need for a contextual, forward-looking, quality-aware decision-support system

SOLUTION

A patient-specific glucose forecasting workspace

KISS integrates historical clinical information with real-time wearable signals to provide a **unified**, forward-looking view of a patient's glucose trajectory, with interactive what-if scenarios for clinical decision support.

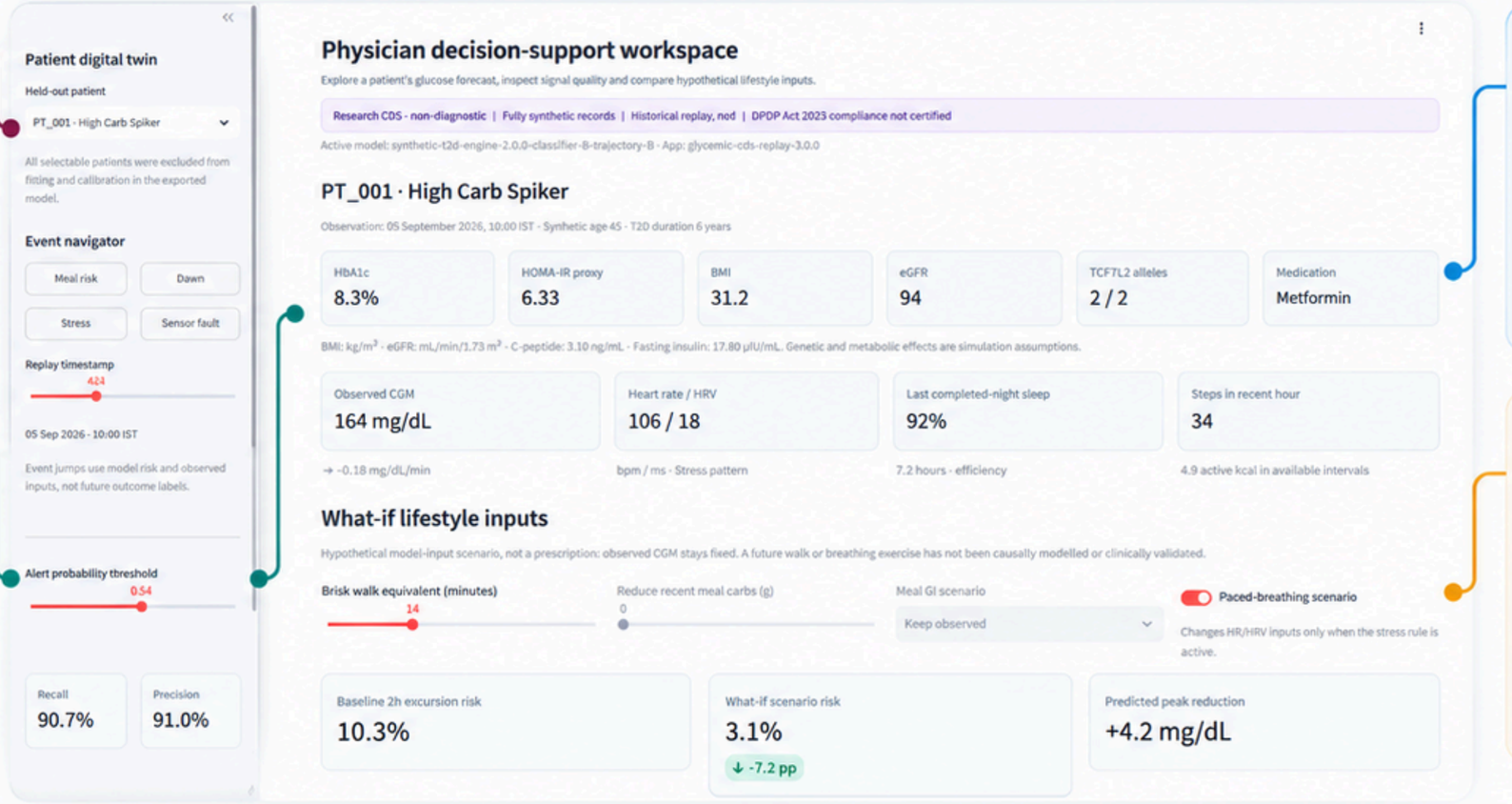
One workspace.
Richer context. Earlier insights.
More informed decisions.

1 Patient context

- Historical clinical profile
- Key metrics (HbA1c, BMI, eGFR, medication, etc.)
- Held-out synthetic patients for unbiased evaluation

2 Signal monitoring

- CGM glucose (15-min)
- Heart rate / HRV
- Activity & steps
- Sleep patterns
- Meal logs



3 Forward forecast

- 2-hour glucose trajectory
- Excursion risk probability
- Predicted peak glucose
- Visual trend with uncertainty band

4 What-if scenarios

- Adjust recent activity
- Reduce meal carbohydrates
- Change meal GI
- Paced breathing (stress)
- See updated risk and peak

HEALTHCARE USE CASE

Detect a potential glucose excursion early

Combining historical clinical data with real-time wearable signals to forecast short-term glucose trends, helping clinicians take timely and personalised action.



1 Establish context
Historical EHR and baseline clinical variables (e.g., HbA1c, BMI, diagnoses, medications)

2 Observe signals
Recent CGM values, trend, meal logs, activity, sleep and heart rate/HRV

3 Forecast
Estimate the next 2 hours of glucose and probability of excursion using the patient's digital twin

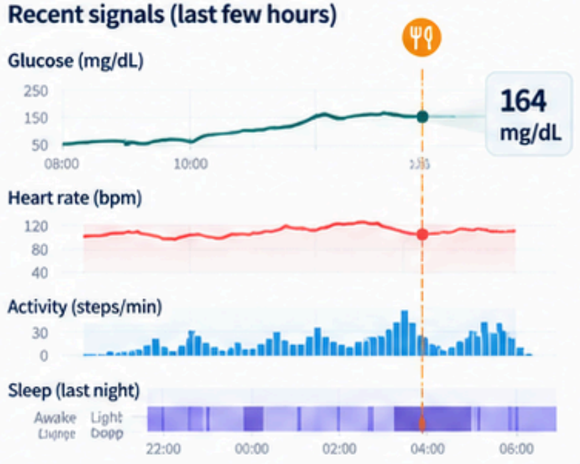
4 Review & act
Clinician inspects forecast, data quality and what-if scenarios to guide decisions

Patient profile (example)

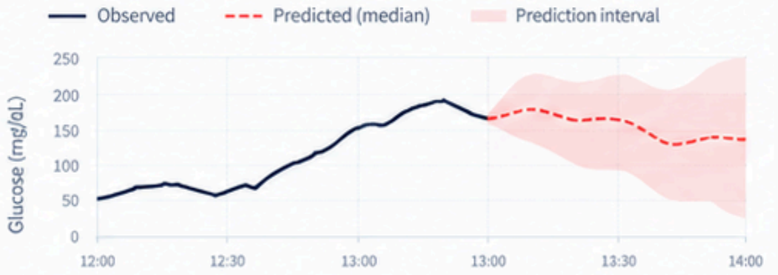
PT_001 · High Carb Spiker

Age	45 years
Gender	Male
BMI	31.2 kg/m²
HbA1c	8.3%
eGFR	94
Medication	Metformin

Recent signals (last few hours)



2-hour glucose forecast



Excursion risk	Predicted peak	What-if scenario risk
10.3%	165.8 mg/dL ↑	3.1% ↓ -7.2 pp

What-if scenario (illustrative)

- Brisk walk equivalent (minutes): 14 (0 – 60)
- Reduce recent meal carbs (g): 40 (0 – 100)
- Meal GI scenario: Lower GI meal
- Paced-breathing scenario: on

Clinical value

- Earlier identification of potential glucose excursions
- Personalised, patient-specific insights
- Supports proactive and informed decisions

ARCHITECTURE

From data to decision support

KISS integrates historical clinical records with real-time wearable signals, processes and validates the data, and uses a machine learning–based digital twin to generate personalised glucose forecasts and insights.

Historical Clinical Data (EHR)

- Demographics (age, gender)
- Clinical history (diagnoses, comorbidities)
- Lab results (HbA1c, C-peptide, eGFR, etc.)
- Medications (e.g., Metformin, Insulin)
- Other risk factors (e.g., TCF7L2)

Wearable / IoT Data

- Continuous glucose (CGM) (every 15 minutes)
- Heart rate / HRV
- Activity & steps
- Sleep patterns
- Meal logs (timing, carbohydrate intake)

Data Processing & Validation

Data cleaning
(missing values, outliers)

Time alignment
(sync EHR and time series)

Feature engineering
(trend, variability, lifestyle features)

Input quality checks
(invalid packets, sensor gaps)

Digital Twin Inference Engine

Risk Classification

- Model A: L2 Logistic Regression (with calibration)
- Model B: Histogram Gradient Boosting (sigmoid calibrated)

Outputs: Probability of glucose excursion

Glucose Trajectory Forecasting

- Multi-output regression (next 2 hours)
- Ridge Regression / Histogram Gradient Boosting

Outputs: Predicted glucose trajectory & peak value

Outputs & Insights

- 2-hour glucose forecast
- Excursion risk probability
- Key contributing factors
- What-if scenario simulation
- Data quality indicators
- Uncertainty band

Physician Dashboard (Decision Support)



- Patient overview
- Interactive what-if scenarios
- Explainable insights
- Audit and transparency

Unified patient view (EHR + real-time signals)

Personalised, forward-looking forecasts

Data quality and uncertainty awareness

Supports proactive and informed clinical decisions

TECH STACK, AI/ML & LICENSING

Built with open tools. Powered by data and ML.

A simple, reproducible and interpretable tech stack to integrate clinical and wearable data, build machine learning models and deliver a physician-facing dashboard.

Open, reproducible and extensible for future use cases.

1 Technology Stack

Programming Language	Python	(3.10+)
Dashboard & UI	Streamlit	(Interactive web app)
Data Processing	Pandas	Data wrangling and analysis
Numerical Computing	NumPy	Numerical operations
Machine Learning	Scikit-learn	Model training and evaluation
Data Storage	CSV files	Synthetic datasets
Visualisation	Matplotlib / Seaborn	Charts and insights

2 AI/ML Models & Framework

A. Risk Classification Model

- Model 1: L2 Logistic Regression (with sigmoid calibration)
- Model 2: Histogram Gradient Boosting Classifier (sigmoid calibrated)

Output

Probability of glucose excursion (in next 2 hours)

B. Glucose Trajectory Forecasting

- Multi-output Ridge Regression
- Histogram Gradient Boosting Regressors (8 horizon steps)

Output

Predicted glucose trajectory & peak value

C. Model Evaluation Framework

- 80:20 split by patient (primary evaluation)
- Separate calibration group for probability calibration

Metrics

AUROC, Recall, Precision, FPR, Trajectory RMSE

3 Open-Source Licensing & Reproducibility

Open-Source Licence

Released under the MIT License (free to use, modify and distribute with attribution).

MIT License

Reproducible Pipeline

Complete code, data processing, model training and dashboard included in the repository.

Data

Synthetic datasets generated for research and evaluation purposes only.

Dependencies

All required Python packages listed in requirements.txt with version information.

Research Prototype

Uses synthetic data. Not for clinical use, diagnosis or treatment. No live device integration.

Open tools and data | Reproducible and extensible | Interpretable models | Designed for real-world healthcare workflows

RESULTS

High predictive performance on synthetic patient data

Our models accurately identify potential glucose excursions and forecast the next 2-hour trajectory using historical clinical data and real-time wearable signals.

Demonstrates the feasibility of a patient-specific digital twin for glucose forecasting.

0.993
AUROC
Model B · 80:20 split

91.1%
Recall (Sensitivity)
Model B · 80:20 split

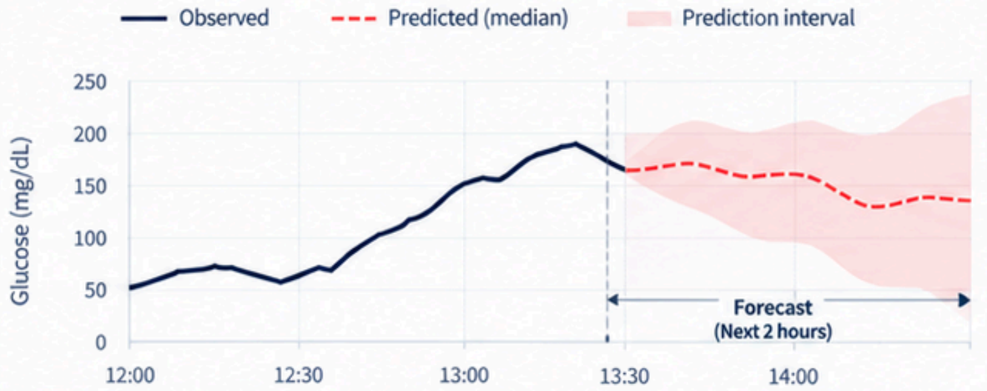
90.4%
Precision (PPV)
Model B · 80:20 split

10.16
Trajectory RMSE
mg/dL · Model B

Model comparison (Test set)

Metric	Model A (L2 LR + Ridge)	Model B (Hist. GB + GB Regressor)
AUROC	0.978	**0.993**
Recall (Sensitivity)	44.1%	**91.1%**
Precision (PPV)	99.8%	**90.4%**
False Positive Rate	0.0%	1.8%
Trajectory RMSE (mg/dL)	9.38	**10.16**

2-hour glucose forecast (example patient)



Key takeaways

1. **High discrimination**
   Model B achieves **0.993** AUROC on synthetic test set.
2. **Balanced performance**
   91.1% recall and **90.4%** precision detect potential excursions effectively.
3. **Accurate forecasting**
   Trajectory RMSE of **10.16** mg/dL over a 2-hour horizon.
4. **Useful for decision support**
   Provides early signals to enable timely, personalised interventions.

Evaluation setup
80:20 split by patient
Separate calibration group

Dataset
Synthetic simulator-derived data
(Research prototype)

Important note
Results are based on synthetic data with 8 test patients.
These scores do not establish real-world clinical performance.

LIMITATIONS & FUTURE WORK

Current limitations and future roadmap

KISS is a research prototype that demonstrates the feasibility of a patient-specific digital twin for glucose forecasting using synthetic data. There are important limitations, and several high-value opportunities for future work.

Current limitations

1. **Synthetic data only**
   Trained and evaluated on simulator-generated data.
   Results do not establish real-world clinical performance.
2. **Small evaluation set**
   Test set includes 8 patients with an 80:20 split.
   Limited demographic and clinical diversity.
3. **No live device integration**
   Uses stored CGM and other signal data.
   Not integrated with real wearable devices or EHR systems.
4. **Narrow use case scope**
   Current prototype focuses on glucose forecasting.
   Cardiovascular and hypertension models are not yet built or validated.
5. **Not a medical device**
   This is a decision-support research prototype, not a diagnostic or treatment system. Clinical validation and regulatory approvals are required for real-world use.

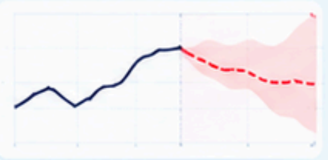
Future roadmap

1. **Validate on real-world data** — 6–12 months
   Evaluate on larger, diverse patient cohorts with real EHR and wearable data.
2. **Live data integration** — 6–12 months
   Integrate with CGM devices and EHR systems for real-time data ingestion and monitoring.
3. **Enhanced models** — 12–18 months
   Explore deep learning and multi-modal models for improved accuracy and robustness.
4. **New use cases** — 12–24 months
   Extend the digital twin to cardiovascular risk forecasting and hypertension management.
5. **Clinical validation & deployment** — 24+ months
   Conduct clinical studies, ensure regulatory compliance and pilot in real-world settings.

Planned expansion use cases

Glucose forecasting (Current prototype)

- 2-hour glucose trajectory
- Excursion risk probability
- What-if scenario simulation



Cardiovascular risk (Future work)

- Heart rate and HRV trends
- Risk of cardiac events
- Lifestyle intervention scenarios



Hypertension management (Future work)

- Blood pressure trend forecasting
- Risk of hypertensive episodes
- Medication and lifestyle scenarios



Key takeaways

- **Proof of concept** — Demonstrates feasibility of a patient-specific digital twin for glucose forecasting.
- **Clear path forward** — Real-world data, device integration and clinical validation are key next steps.
- **High potential** — Can be extended to cardiovascular and hypertension use cases.
- **Responsible innovation** — Designed to augment, not replace, clinical judgement.

Thank You

Patient Digital Twin

From fragmented data to earlier insights, for more informed and personalised care.



Brings data together

Combines historical records with real-time signals

Anticipates risks

Forecasts potential glucose excursions before they occur

Supports clinicians

Provides explainable, patient-specific insights

Explores a healthier future

Enables what-if scenarios for timely, personalised action

A step towards proactive, personalised and data-driven healthcare.

Keep It Simple, Stupid.

Using data and technology to make better health decisions, one patient at a time.

Team KISS
Shivam Soni (Team Leader) and Somesh | IIM Jammu